SQL Data Analysis Internship

TASK – 1

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JUNE BATCH

Objective

To perform database operations using SQL and analyze the stored data using various queries such as aggregate functions, grouping, sorting, and filtering. The results obtained from these queries are presented and explained in this report.

1) SELECT * FROM Students;
   Shows the details of all the students.

StudentID	Name	Gender	Age	Grade	MathScore	ScienceScore	EnglishScore
1	Abha	F	20	B	78	79	80
2	Riya	F	22	A	90	92	88
3	Akshit	M	21	B	77	80	80
4	Dolma	F	20	D	60	60	58
5	Bhanu	F	21	A	86	82	85
6	Ram	M	20	A	90	90	93
7	Bharat	M	20	C	60	68	75
8	Raj	M	23	B	75	80	79
9	Jenny	F	19	F	30	33	40
10	Kate	M	20	E	55	50	52
11	Heena	F	20	A	90	92	89

Figure 1

2) SELECT AVG(MathScore),

   AVG(ScienceScore),

   AVG(EnglishScore)

   FROM Students;

This query finds the average score of all students in Mathematics, Science, and English. The AVG() function is used to calculate the mean marks for each subject.

AVG(MathScore)	AVG(ScienceScore)	AVG(EnglishScore)
71.9091	73.2727	74.4545

Figure 2

3) SELECT StudentID,name,

   (MathScore+ScienceScore+EnglishScore) AS total_marks

   FROM Students

   ORDER BY total_marks DESC LIMIT 1;

This query identifies the overall top-performing student by calculating the total marks obtained in Mathematics, Science, and English. The results are sorted in descending order of total marks using ORDER BY, and LIMIT 1 returns the student with the highest total score.

	StudentID	name	total_marks
▶	6	Ram	273

Figure 3

4) SELECT COUNT(StudentID), Grade

   FROM Students

   GROUP BY (Grade);

This query counts the number of students in each grade category. The COUNT(StudentID) function calculates how many students belong to each grade, and the GROUP BY Grade clause groups students according to their grade before counting them.

	COUNT(StudentID)	Grade
	4	A
	1	D
	1	C
	1	F
	1	E

Figure 4

5) SELECT gender,

   AVG(MathScore),

   AVG(ScienceScore),

   AVG(EnglishScore)

   FROM Students

   GROUP BY (gender);

This query calculates the average marks obtained by male and female students in Mathematics, Science, and English separately. The GROUP BY gender clause groups students according to their gender, and the AVG() function computes the average score for each subject within each gender group

gender	AVG(MathScore)	AVG(ScienceScore)	AVG(EnglishScore)
F	72.3333	73.0000	73.3333
M	71.4000	73.6000	75.8000

Figure 5

6) SELECT StudentID,Name,MathScore

   FROM Students

   WHERE MathScore > 80;

   This query retrieves the Student ID, Name, and Mathematics score of students who scored more than 80 marks in Mathematics. The WHERE clause is used to filter the records and display only those students whose MathScore is greater than 80

StudentID	Name	MathScore
2	Riya	90
5	Bhanu	86
6	Ram	90
11	Heena	90
NULL	NULL	NULL

Figure 6

7) UPDATE Students

   SET MathScore=75

   WHERE StudentID = 1;

   This query changes the Mathematics score of Student ID 1 from 78 to 75 using the UPDATE statement. The WHERE clause is used to update only the selected student's record.